introduced into a tower or chamber along with heated air. The small droplets come into contact with the hot air, blast off their moisture, become small particles and drop to the bottom where they are removed. This method can produce a high quality product even with heat sensitive products like milk, eggs and coffee.

- **Drum or Roller drier** – Liquid foods, purees and mashes are dried by this method. The food to be dried is applied, as a continuous thin layer, on to the surface of a revolving drum or between a pair of drums moving in opposite directions heated by steam. The dried layer of food is scraped by a scraper blade positioned at a point on the drum. Foods that are sticky cannot be scraped when it is hot. Such a sticky food becomes brittle when cooled, which facilitates scraping. For heat resistant food products, drum drying is one of the inexpensive dehydration methods.
- **Vacuum driers** – This method is quiet expensive but gives good quality foods. It consists of a vacuum chamber that can withstand air pressure and contains shelves to hold food. The shelves are heated. The food gets heated by conduction and radiated heat. Liquid foods dehydrated by vacuum drying have a puffed structure and are easily dissolved in water. There is minimum flavour change and heat damage because low temperature is used in this method.

Dried foods are very convenient as they are light weight, take up little storage space and can be stored for long periods as emergency foods.

4.2.4 Smoking of Foods

Smoking is mostly done to preserve the meat. This process helps to develop flavours in it. Wood smoke contains small amounts of formaldehyde, higher aldehydes, formic acid, acetic acid and resins. These compounds have antiseptic properties and destroy microorganisms present. The temperature and period of smoking vary with the type of meat. In sausages, the smoking is done for a few hours after smoking the material is packed in polythene bags and kept at refrigerated conditions.

4.2.5 Preservation by Chemical Preservatives

Preservatives are chemical agents which serve to retard, hinder or mask undesirable change in food. Preservatives help in retaining the original quality of food and delaying their spoilage.

Preservatives are classified into class I and class II preservative. Class I preservatives are available at home. Class II preservatives are prepared in the industries.

- Sulphur dioxide is the only permitted preservative used in the form of sulphites.
- In India, sodium benzoate, sulphites and sorbic acids are permitted preservatives used in fruits and vegetables.
- Dried fruits are treated with sulphur dioxide to conserve the colour and to prevent the growth of microorganisms.
- Sodium benzoate is preferred to benzoic acid because of its solubility and used in tomato ketchups, sauces, jams, jellies, pickles and fruit juices.

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